

# Introduction to data science for wind energy systems

## Exercise: Large-scale wind generation time series

Data for the exercise are given in four files (in subfolder “Data”):

- “Wind\_generation\_simulated.csv”: One year of simulated onshore wind generation data for 3 countries
- “Solar\_generation\_simulated.csv”: One year of simulated solar generation data for the same 3 countries
- “Wind\_generation\_simulated\_model2.csv”: One year of simulated onshore wind generation data (only 1 country), using a different simulation model (model 2)
- “Wind\_generation\_measured.csv”: One year of measured onshore wind generation data (only 1 country)

All data are given in GMT time zone. All data are given as standardized generation, i.e., as share of installed capacity (0 means no generation and 1 means generation at full installed capacity).

### Exercise 1: Wind generation in multiple countries

Load the simulated wind generation data for the three countries.

**1.1:** Visualize the data as time series, on hourly resolution, daily resolution (daily mean) and monthly resolution (monthly mean).

**ANSWER:** See the results from running [Exercise\\_large\\_scale\\_time\\_series\\_KEY.ipynb](#)

**1.2:** Visualize the data also as PDFs (hourly data). You can use different methods (e.g., histogram based or kernel density estimation). Which PDF visualization method appears the most appropriate for the data? Why?

**ANSWER:** See the results from running [Exercise\\_large\\_scale\\_time\\_series\\_KEY.ipynb](#). As the wind generation data are bounded between 0 and 1 (1 = generation at installed capacity), this information should be provided to kernel density estimation type methods (which do smoothing). (see [Exercise\\_large\\_scale\\_time\\_series\\_KEY.ipynb](#) for how this can be done). Histogram-based PDF estimation is a simple and transparent method which generally works well with wind and solar generation data.

**1.3:** Calculate mean, SD and relative SD (RSD) for each country (hourly data). Which country has the highest RSD? Can you think of any reason why? Which country has the lowest RSD? (RSD = SD divided by mean)

**ANSWER:** See the results from running [Exercise\\_large\\_scale\\_time\\_series\\_KEY.ipynb](#). Denmark (DK) has the highest RSD (indicating most variability), which seems reasonable as Denmark is geographical smallest of the three analyzed countries (i.e., it has less geographical smoothing). Finland (FI) has the lowest RSD, even though it shows the highest SD. The high mean generation in Finland, driven by the most modern wind power plant fleet of the three analyzed countries, lowers the RSD. The fact that RSD

is impacted both by SD and the mean generation makes it often a more suitable metric than SD for comparing variability across technologies with very different mean generation.

**1.4:** Calculate aggregate onshore wind generation of the three countries, assuming each country has the same amount of installed capacity (aggregate generation is the sum of the generation at each time step; however, note that also the aggregate generation should be given as standardized generation). Calculate the same visualizations and descriptive statistics for the aggregate generation as for the individual countries in 1.1-1.3. How does the PDF of the aggregate generation look compared to the individual countries? Compare the RSD of the aggregate generation to the RSDs of the individual countries. How does it compare? Why?

**ANSWER:** See the results from running `Exercise_large_scale_time_series_KEY.ipynb`. PDF of the aggregate generation shows generally lower probabilities for very low or high generation compared to the PDFs of the individual countries. This is driven by geographical smoothing. This is also evident in the RSD, where the aggregate generation shows lower RSD than any individual country.

**1.5:** Calculate hourly generation ramp time series for the individual countries and for the aggregate generation. Visualize the regeneration ramp PDFs for the individual countries and for the aggregate and calculate respective the ramp SDs. How do the aggregate ramps look compared to the individual countries? Why?

**ANSWER:** See the results from running `Exercise_large_scale_time_series_KEY.ipynb`. The generation ramp distribution of the aggregate shows lower probabilities for large negative or positive ramps. This is evident also in the generation ramp SD, where the aggregate shows significantly lower SD than any individual country. (in this case where the mean is practically zero, the use of RSD makes less sense)

## **Exercise 2: Wind and solar generation in multiple countries**

Keep the data from Exercise 1 open and load also the simulated solar generation data for the three countries.

**2.1:** Visualize the solar generation data as time series, on hourly resolution, daily resolution (daily mean) and monthly resolution (monthly mean). How does it look compared to wind generation? Why?

**ANSWER:** See the results from running `Exercise_large_scale_time_series_KEY.ipynb`. Solar generation shows higher generation during late Spring and summer, whereas wind generation shows higher generation during winter. This indicates complementarity between the two VRE sources.

**2.2:** Calculate correlations (Pearson correlation) between all the wind generations. What can you say about the correlations? Why are some low and some high?

**ANSWER:** See the results from running `Exercise_large_scale_time_series_KEY.ipynb`. Correlation between Danish and German (DE) wind generation is relatively high (0.68), whereas correlations between Finland and Denmark and Germany are significantly lower ( $< 0.35$ ). This is driven by the geographical distances between the different countries.

**2.3:** Calculate correlations (Pearson correlation) between all the solar generations. What can you say about the correlations? How do they appear compared to the wind generation correlations? Why do you think that is?

**ANSWER:** See the results from running `Exercise_large_scale_time_series_KEY.ipynb`. Correlations between the solar generations in the three countries are all quite high ( $> 0.79$ ). There is some impact of geographical distance, as the correlation between DK and DE is the highest (0.88), but overall all correlations are high due to the highly correlated diurnal (day/night) and seasonal (summer/winter) rhythms between the different countries.

**2.4:** Calculate correlations (Pearson correlation) between all the wind and the solar generations. What can you say about the correlations between wind and solar generation?

**ANSWER:** See the results from running `Exercise_large_scale_time_series_KEY.ipynb`. All wind-solar generation correlations are negative. This indicates complementarity between solar and wind generation across the different countries.

**2.5:** Calculate aggregate onshore wind and solar generation of the three countries, assuming each country has the same amount of installed capacity for both wind and solar (give the aggregate generation as standardized generation). Compare the RSD of this aggregate wind and solar generation to the aggregate wind generation from 1.5. How does it compare?

**ANSWER:** See the results from running `Exercise_large_scale_time_series_KEY.ipynb`. RSD of the aggregate considering both wind and solar generation is lower than RSD of only considering wind generation (see 1.5). While the wind and solar generation installations are not optimized in these exercises, this indicates that there is at least potential for reducing aggregate VRE generation variability by having a mixture of wind and solar generation (which is logical considering the negative correlation between wind and solar generation).

### Exercise 3: Compare and validate wind simulation models

Keep the data from Exercise 1 open and load also the simulated wind generation data using a different model (model 2). Load also the measured wind generation data.

**3.1:** Plot the two simulations models and the measured for Finland, on hourly resolution, daily resolution (daily mean) and monthly resolution (monthly mean). How well would you say the two models represent the measured data?

**ANSWER:** See the results from running `Exercise_large_scale_time_series_KEY.ipynb`. Overall, both models seem to capture at least daily and monthly level generation quite well. Even the hourly level seems reasonably well capture. From looking at the time series plots, neither models seems obviously better.

**3.2:** Compare the two models regarding validation to the measured data in terms of RMSE (hourly data). Which model looks better?

**ANSWER:** See the results from running `Exercise_large_scale_time_series_KEY.ipynb`. Considering RMSE, model 1 looks better (lower RMSE).

**3.3:** Compare the two models regarding validation to the measured data in terms of distributional properties (look at the PDFs, calculate EMD metrics, look at very low/high generation value likelihoods, etc.). Which model looks better?

**ANSWER:** See the results from running [Exercise\\_large\\_scale\\_time\\_series\\_KEY.ipynb](#). Considering EMD, model 1 looks better (lower EMD). Looking at the PDF seems to roughly back this up. However, the very maximum simulated generation from model 2 appear closer to measured maximum.

**3.4:** Compare the two models regarding correlation to the measured data. Which model looks better?

**ANSWER:** See the results from running [Exercise\\_large\\_scale\\_time\\_series\\_KEY.ipynb](#). Model 1 looks better (higher correlation to measured data).

**3.5:** Compare the two models in getting the annual production similar to the measured data. Which model looks better?

**ANSWER:** See the results from running [Exercise\\_large\\_scale\\_time\\_series\\_KEY.ipynb](#). Model 2 shows a lower “Mean\_err”, i.e., error in the mean annual generation.